

George Mamaladze

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Currently

- **Senior Software Architect** at *Siemens AG, Factory Automation, HMI*, Munich, Germany
- **Chief Technical Advisor** at *Syniotec*, Bremen, Germany (remote)

Summary

Rare combination of deep conceptual technical knowledge and hands-on expertise. Technology leader with over 25 years of experience designing, building, and delivering software products and solutions. Ability to build and lead high-performing teams, organize processes, and drive innovation. Experience in interfacing with customers, understanding requirements, collaborating with executive leadership, developing and implementing strategies.

EXPERIENCE

Senior Software Architect (full-time, employed)

May 2022 - Present

Siemens AG – Factory Automation, HMI, Munich, Germany

The SIMATIC WinCC Unified System is a new generation of Siemens HMI and SCADA visualization products designed for industrial applications in machine building and factory automation. It addresses the modern challenges of digitization in manufacturing by providing a flexible and open platform with IT/OT integration capabilities.

www.siemens.com/wincc-unified

Responsibilities:

- Defining technology strategy, aligning product roadmap with business goals and partners
- Led and mentored the engineering team Architects
- Drove the technology roadmap, scanning for innovations, defining development guidelines and tool chains
- Oversaw the entire development lifecycle, allocating technical resources, collaborating with product owners, finance, and executive leadership to ensure timely delivery and market fit

Key Achievement:

- After 3 years of development of the newly founded business line, delivered the first version of the product to market (currently in limited release, broad release planned)

Chief Technical Advisor (part-time, freelance)

Jul 2022 - Present

Syniotec, Bremen, Germany (remote)

Syniotec is a fast-growing startup providing digital solutions to the construction industry, enabling companies to optimize operations through IoT-driven telematics and software platforms. As Chief Technical Advisor, I function in practice as the part-time CTO, shaping product vision, technical strategy, and long-term platform strategy.

www.syniotec.com

Responsibilities:

- Defined the technical roadmap ensuring scalability, security, and maintainability.
- Advised the founding team and management.
- Mentored engineering teams on best practices in microservices, DevOps, and cloud infrastructure.
- Managed technology stack, vendor partnerships, and integration strategies.

Key Achievements:

- Established technical standards and processes that enabled Syniotec's transition from startup to scaling vendor.
- Redesigned the cloud-native backbone of Syniotec software to scale for thousands of concurrent users and data streams.

Technical Trainer & Consultant (part-time, freelance)

Feb 2020 - Jun 2022

Independent, Germany

Provided training and consulting services to multiple software companies, focusing on modern cloud-native architecture and microservices implementation.

Responsibilities:

- Created and delivered training programs on microservices architecture and cloud-native development: Microservice Patterns, Microservices implementation in Quarkus vs. Spring Boot, Testing strategies for microservices
- Conducted architecture review workshops

Key Achievement:

- Trained over 100 developers across multiple companies, overwhelming positive feedback and measurable improvements in project outcomes.

Senior Software Architect (full-time, employed)

May 2018 - May 2022

Siemens AG – Corporate Technology, Munich, Germany

Siemens Corporate Technology served as the central research and innovation hub for Siemens AG.

www.siemens.com/corporate-technology

Responsibilities:

- Designed and conceived cloud-native system architectures for strategic projects.
- Advised business units on cloud transformation and innovation.
- Performed architecture reviews and mentored development teams.
- Conducted technology research, academic collaboration, and conference talks.

Key Achievements:

- Led the architecture for Spectrum Power digital grid control system (real-time platform with Kubernetes, Kafka, Flink).
- Architected "Energy as a Service," a scalable microservice platform for renewable energy analytics.

Advisor for Small Businesses (part-time, freelance)

Oct 2018 - Jan 2020

EBRD, Global

EBRD (European Bank for Reconstruction and Development) Business Advisory Service programme supports small and medium-sized companies in emerging markets.

www.ebrd.com/home/what-we-do/products-and-services/support-for-start-ups-and-msmes/access-business-advice.html

Responsibilities:

- Provided strategic guidance and coaching to small businesses in Eastern Europe & CIS
- Mentored software teams in modern engineering practices and agile methods
- Reworked system architectures to align with business objectives
- Introduced structured processes, tools, and templates to improve execution

Key Achievement:

- Overwhelmingly positive feedback from clients backed by measurable business success.

Principal Software Architect (full-time, employed)

Mar 2017 - Apr 2018

Huawei – European Research Center, Munich, Germany

Huawei European Research Center focuses on cutting-edge research and innovation to introduce these into Huawei's products and solutions globally.

www.huawei.com

Responsibilities:

- Provided technical leadership for 3-5 concurrent high-impact research projects
- Drove project acquisition, requirements analysis, and technical consulting at HQ in China
- Disseminated research findings and knowledge

Key Achievements:

- Architected methodology and toolchain for modularizing large-scale Java projects.
- Designed scalable microservice architecture for Huawei Cloud IDE.

Senior Software Architect (full-time, employed)

Jul 2009 - Feb 2017

Siemens AG – Digital Factory, Fuerth, Germany

Started as Software Architect, promoted to Senior Software Architect in 2015. Worked on TIA-Portal, Siemens' flagship automation IDE.

www.siemens.com/tia-portal

Responsibilities:

- Created and implemented architecture of platform components across full stack (hardware, firmware, application).
- Interfaced with customers and partners for product launches
- Built highly complex products using Product Line Engineering approaches

Key Achievements:

- Shipping 5 major releases.
- Introduced DevOps practices, CI/CD pipelines, and automated testing frameworks
- Spearheaded Test Driven Development (TDD) adoption across multiple teams
- Championed Agile methodologies, measurably increasing delivery speed and product quality

Senior Software Architect (full-time, employed)

Feb 2008 - Jun 2009

encad Ing. mbH, Nuremberg, Germany

Encad was a Siemens subcontractor providing staff leasing services. Worked on Siemens TIA-Portal project as Software Architect.

Responsibilities:

- Assigned to a customer (Siemens) project with 50+ developers
- Conducted architecture reviews and systemic root cause analyses
- Introduced CI, static code analysis, and quality assurance measures

Key Achievement:

- Excellent satisfaction feedback from Siemens, which led to transition to Siemens employment.

Software Project Lead (full-time, employed)

Aug 2001 - Jan 2008

IQ-optimize Software AG, Nuremberg, Germany

IQ-Work Software AG was subsidiary of Drillisch AG, Germany's leading mobile telecommunications distributor/retailer.

www.iq-optimize.com

Responsibilities:

- Architecting and leading development of a digital prepaid voucher distribution system
- Managed lifecycle of a back-office workflow system (product ownership, development, customer training)

Key Achievements:

- Launched Germany's first digital prepaid voucher distribution system with a 10-person team
- Delivered and introduced back-office workflow system to multiple customers

Head of IT Department (full-time, employed)

Jan 1999 - Jul 2001

TBC Bank, Tbilisi, Georgia

TBC Bank is one of the largest banks in Georgia, and one of the leading financial institutions in the region.

www.tbcbank.ge

Responsibilities:

- Managed a team of IT professionals
- Oversaw administration and supervision of IT infrastructure hardware, software, networks, ATMs, POS systems, and integrations

Key Achievements:

- Created first data center for TBC Bank.
- Introduced new state of the art core banking software.
- Introduced SWIFT connectivity, improving international transaction capabilities.
- Integrated into Visa and MasterCard networks.
- Installed first ATM machines in Georgia.

EDUCATION**Post-graduate Studies in Computer Science**

Sep 2003 - Aug 2004

FernUni Hagen, Hagen, Germany

- Free student (Gasthörer), completed 4 semesters

Diploma in Computer Science

Sep 1993 - Jun 1999

Tbilisi State University, Tbilisi, Georgia

- Faculty of Applied Mathematics and Computer Science
- Scholarship holder of the George Soros Foundation

Specialized School for Physics and Mathematics

Sep 1989 - May 1993

High School, Tbilisi, Georgia

- Graduated with honors
- Winner (1st Place), International Young Physicists' Tournament

ACHIEVEMENTS

- **2018** - Certified AWS Solution Architect
- **2015** - CodeProject MVP
- **2014** - Certified Senior Software Architect
- **2013** - Member of Agile Alliance
- **2013** - Agile Manifesto Translator
- **2013** - Co-author of several patents

LANGUAGES

- **English** - Fluent
- **German** - Fluent
- **Chinese** - Basic
- **Russian** - Native
- **Georgian** - Native

TECHNICAL SKILLS

Languages & Frameworks: C# • Java • Python • TypeScript • Node.js • ASP.NET Core • Angular

Architecture: OO-Patterns (GoF) • Microservice Patterns • Architecture Evaluation • Requirements Engineering • Domain Driven Design • Testing Strategies • Guidance & Governance • Continuous Delivery • AWS Well-Architected Framework • Cloud Cost Management • Event-Driven Architecture

Platforms & Products: Cloud Platforms • Kubernetes • Kafka • Flink • PostgreSQL • DynamoDB • Terraform • Prometheus • Grafana • OpenTelemetry

Tools: Git • Azure DevOps • Visual Studio Code • IntelliJ IDEA • Docker • GitHub Actions • GitLab CI • SonarQube • JFrog

Practices: Agile • Scrum • Lean Management • Clean Code • TDD • CI/CD • Management 3.0